

Choosing How to Extend the Ladder over Time

If a time segmentation approach does not offer clear rules about the method for extending the bond ladder over time, then it is not a true retirement income plan that can be tested and analyzed.

I will investigate examples of three different methods for choosing when to extend the bond ladder as retirement progresses. These are based on the three methods for creating rolling bond ladders mentioned earlier in Exhibit 7.10: (1) automatic, (2) market-based, and (3) personalized.

Automatic rolling ladders, which Huxley and Burns call “rolling horizon,” keep the same time horizon perpetually by automatically rolling out the ladder length each year as a bond matures. This is done for as long as the growth portfolio has sufficient assets to extend the ladder length. An initial ladder length is chosen, and the ladder is built. At the end of each year, the ladder is extended by one year so that the ladder length remains fixed over time. If the growth portfolio is 100 percent stocks, then this strategy involves taking ongoing distributions from a 100 percent stock portfolio. The only recourse is that the distribution amount is not the full value of spending, but rather the discounted value of the bonds to be purchased based on current interest rates. If and when the growth portfolio depletes, the ladder will continue to provide income without being further extended until it is wound down completely and all assets are depleted.

Exhibit 7.14 provides the outcomes for one Monte Carlo simulation for stock returns and bond yields to show how an automatic rolling ladder might perform in practice. This retiree seeks to support an initial \$40,000 spending goal with an annual 2 percent cost-of-living adjustment for as long as possible in retirement. The retiree has \$1 million at the start of retirement to divide between a bond ladder and a stock portfolio. The retiree seeks to maintain a ten-year bond ladder providing the desired annual spending throughout retirement. With a 2.45 percent initial bond yield and assuming a flat yield curve, building the initial bond ladder requires 39.2 percent of the asset base. The other 60.8 percent of assets are left in stocks. While I allow interest rates to fluctuate randomly over time, I do simplify by assuming a flat yield curve. Relative to an upward sloping yield curve, a flat yield curve lowers the costs of short-term spending and raises the costs of long-term spending. But these factors will tend to offset one another, and with low